

Job Scout: An AI-Driven Resume Parsing and Job Matching Platform

Abstract

Today's job market is becoming increasingly competitive due to a growing number of applicants and an over-dependence on bot-driven systems for screening resumes. This project introduces Job Scout, a smart, AI-enabled service that matches potential candidates with suitable job positions through advanced methods known as Retrieval-Augmented Generation (RAG) and mixed vector searches. The platform uses Next.js technology on the front end and FastAPI system on the back end to perform data processing with the aid of PostgreSQL database and pgvector libraries. The service employs the use of an automated web scraper to fetch information about jobs in real time and takes advantage of a Local Large Language Model (LLM) that acquires data from resumes, generates complex vector projections of applicants, and produces relevant documents like cover letters. The platform efficiently provides customized job suggestions to users, sorts job vacancies with the help of multidimensional skills matrix, and visualizes users' skills via a radar diagram.

Keywords: Retrieval-Augmented Generation, Resume Parsing, Artificial Intelligence, Job Matching, Vector Search.

Architecture Diagram

